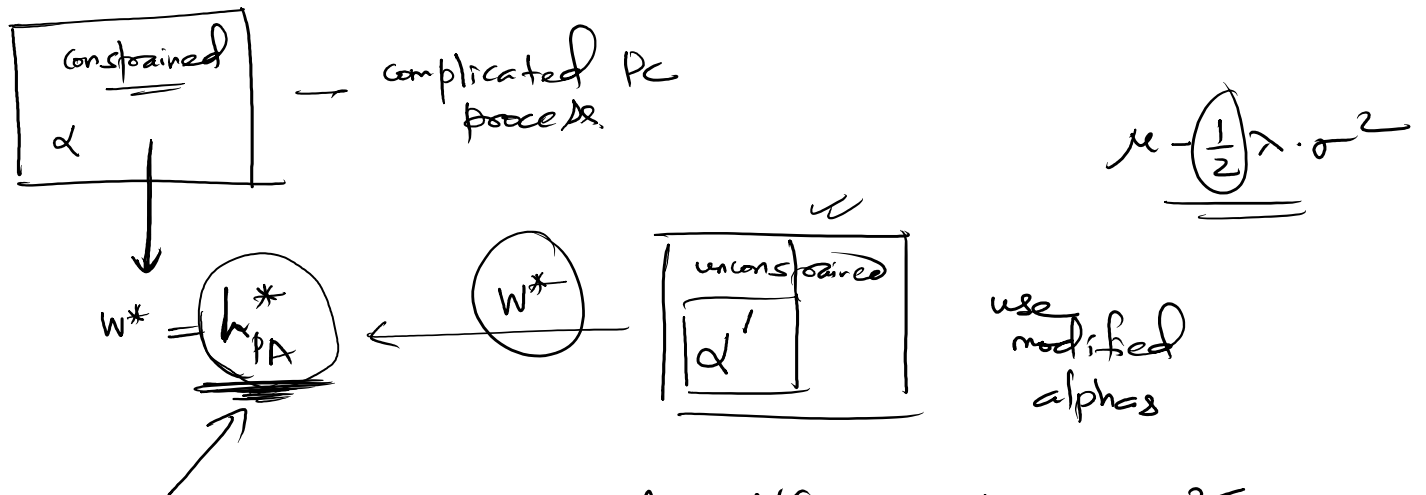




$$\max_w U(w) = \alpha_p - \lambda \cdot \omega_p^2$$

coeff.
 for version.

$$\alpha_p = w^T \alpha' = (w_1 \alpha'_1 + w_2 \alpha'_2)$$

$$\begin{pmatrix} w_1 \\ w_2 \end{pmatrix}^T \begin{pmatrix} \alpha'_1 \\ \alpha'_2 \end{pmatrix}$$

$$\omega_p = \sqrt{w^T \Omega w}$$

covariance matrix
(active terms)

$$U(w) = \alpha_p - \lambda \omega_p^2$$

$$= w^T \alpha' - \lambda (w^T \Omega w)$$

$$\frac{\partial U}{\partial w} = 0, \quad \alpha' - 2\lambda \Omega w^* = 0$$

$$\alpha' = 2\lambda \cdot \Omega w^* = 2\lambda \Omega h_{PA}^*$$

$$\alpha' = \left(\frac{1 R_p}{\omega_p^*} \right) \Omega h_{PA}^*$$

$$\underline{U(\omega_p)} = \underbrace{(1R_p)}_{\omega_p} \omega_p - (\lambda) \omega_p^2$$

$$\underline{1R_p} = \frac{\omega_p \leftarrow}{\omega_p}$$

$$\frac{dU}{d\omega_p} = (1R_p) - 2\lambda \cdot \omega_p^* = 0$$

$$\omega_p^* = \frac{1R_p}{2\lambda}$$

$$U^* = 1R_p \cdot \frac{1R_p}{2\lambda} - \lambda \left(\frac{1R_p}{2\lambda} \right)^2 = \frac{1R_p^2}{4\lambda}$$

$$\underline{\underline{\theta_n(t)}} = \underline{\underline{b}} \underline{\underline{g}} + \epsilon$$

$\theta \rightarrow$ residual return
 $g \rightarrow$ signal

$$\mathbb{E}(\theta_n) = b \mathbb{E}(g)$$

$$\mathbb{E}[\theta_n | g] = b g$$

$$\underline{\underline{\mathbb{E}[\theta | g]}} - \underline{\underline{\mathbb{E}[\theta]}} = \underline{\underline{b}} \underset{\uparrow}{\overset{\downarrow 0}{[g - \mathbb{E}(g)]}}$$

$$\alpha = \frac{\text{Cov}(\theta, g)}{\text{Var}(g)} [g - \mathbb{E}(g)]$$

$$\alpha = \underline{\underline{\text{Cor}(\theta, g)}} \cdot \underline{\underline{\text{stdev}(\theta)}} \cdot \left[\frac{g - \mathbb{E}(g)}{\underline{\underline{\text{stdev}(g)}}} \right]$$

$$= \underline{\underline{\text{IC}}} \times \underline{\underline{\text{volatility}}} \times \underline{\underline{Z}}$$

$\downarrow$
 (information coefficient)

$$\begin{aligned} \beta &= \frac{\text{Cov}(x, y)}{\text{Var}(x)} \\ &= \rho(x, y) \frac{\sigma_y}{\sigma_x} \\ &= \rho(x, y) \frac{\sigma_y}{\frac{\sigma_x}{\frac{x - \mu}{\sigma}}} \end{aligned}$$

$$\underline{\alpha_B} = \sum w_i \alpha_i$$

$$\alpha_i \leftarrow \alpha_i - \beta_i (\alpha_B)$$

$$\begin{aligned} & \sum w_i (\alpha_i - \beta_i \alpha_B) \\ &= \sum w_i \alpha_i - \alpha_B \sum w_i \beta_i \\ &= \alpha_B - \alpha_B = 0 \end{aligned}$$

$$r_1 = \alpha_1 + \beta_1 r_B + \epsilon_1$$

$$r_2 = \alpha_2 + \beta_2 r_B + \epsilon_2$$

$$\begin{aligned} \underline{\bar{r}} &= w_1 r_1 + w_2 r_2 \\ &= \underline{(w_1 \alpha_1 + w_2 \alpha_2)} \\ &\quad + \underline{(w_1 \beta_1 + w_2 \beta_2)} r_B + \tilde{\epsilon} \end{aligned}$$

$$= (\alpha_p) + (\beta_p) r_B + \tilde{\epsilon}$$

$$\sum w_i \beta_i = 1$$

Stock 1, \$100

0	6	12	18	24
0.75	1.5	1.5	1.5	0.75

$TC = \$6$ (over a period of 2 years)

$= \$3$ (annual) $\rightarrow$ 3%

Sub of thumb,

$$\begin{aligned} \underline{TC_{\text{annual}}} &= (\text{round trip cost}) \times (\text{frequency per year}) \\ &= (1.5) \times 2 = \underline{\underline{\$3}} \end{aligned}$$

Profit over 2 years = \$8.

Profit annual = \$4

$TC_{\text{Adjusted}} = 1\%$

Stock 2. (\$100 $\rightarrow$ \$108)

Profit (annual) = 4%

$TC = (1.5) \times \frac{1}{2} = \0.75 , 0.75%.

$TC_{\text{adjusted annual return}} = \underline{\underline{3.5\%}}$

$$\left\{ \begin{array}{c} \underline{N_1} \\ \alpha_i \end{array} \right\} \rightarrow \text{Forecast is available}$$

$$\left\{ \begin{array}{c} \underline{N_0} \\ \alpha_i = 0 \end{array} \right\} \rightarrow \text{no forecast available}$$

$$\alpha'_i \leftarrow \alpha_i - \left(\frac{\sum_{i \in N_1} w_i \alpha_i}{\sum_{i \in N_1} w_i} \right)$$

$$\alpha_B = \sum_{i \in N_1} w_i \alpha'_i + \sum_{i \in N_0} w_i \alpha'_i$$

$$= \sum_{i \in N_1} w_i \left(\alpha_i - \frac{\sum w_i \alpha_i}{\sum w_i} \right) + \sum_{i \in N_0} w_i \alpha'_i$$

$$= \cancel{\sum_{i \in N_1} w_i \alpha_i} - \cancel{\left(\sum_{i \in N_1} w_i \alpha_i \right)} + 0 = 0$$

$$VA = \underline{\alpha_p - \lambda \omega_p^2}, \quad \underline{\alpha_p = \alpha_1 w_1 + \alpha_2 w_2}$$

$$\underline{\frac{\partial(VA)}{\partial w_1}} = \alpha_1 - \lambda \cdot 2\omega_p \cdot \left(\underline{\frac{\partial \omega_p}{\partial w_1}} \right)$$

$$\underline{MCVA}_1 = \underline{\alpha_1 - 2\lambda \omega_p \cdot MCAR}_1$$

$$MCVA_n = \alpha_n - 2\lambda \omega_p \cdot MCAR_n$$

① buy 1 unit of asset n?

$$\Delta V_A = \frac{\partial(VA)}{\partial \omega_n} \times (\Delta \omega_n) = \frac{\partial(VA)}{\partial \omega_n} = MCVA_n$$

$$\underline{MCVA_n} \geq \underline{PC_n}$$

② sell 1 unit of asset n?

$$\Delta V_A = \frac{\partial(VA)}{\partial \omega_n} \times (\Delta \omega_n) = -MCVA_n$$

$$-MCVA_n \geq SC_n$$

$$\Rightarrow \underline{MCVA_n} \leq -SC_n$$

No trade condition

$$-SC_n \leq \underline{MCVA_n} \leq PC_n$$

$$-SC_n \leq \alpha_n - 2\lambda \omega_p \cdot MCAR_n \leq PC_n$$

$$-SC_n + 2\lambda\omega_p \cdot MCAR_n \leq \alpha_n \leq PC_n + 2\lambda\omega_p \cdot MCAR_n$$

(no-trade)